

Separating real Kazakh–Russian code-switching from loanwords

A hand-labeled LID benchmark and an XLM-R filter for шала-қазақша in Kazakhstani user text

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Code: github.com/naadgob/KazNLP · Weights & data: huggingface.co/datasets/naadgob/kaznlp-weights

THE PROBLEM

Off-the-shelf detectors and keyword heuristics call a Kazakh sentence "code-switched" the moment it holds a common Russian loanword. On 241,576 Telegram comments, a narrow Kazakh-plus-Russian heuristic flagged **27,628 as mixed**; reading them by hand, only **460 were genuine code-switching** (about 1.7%). Kazakh written with everyday Russian loanwords swamps the real шала-қазақша, so any corpus count or sentiment number built on automatic LID comes out inflated.

WHAT I BUILT

A document-level LID setup that treats *mixed* as its own class and gets scored against people, not synthetic text. The gold set has **3,076 messages** (1,077 mixed, 1,000 ru, 999 kz) pulled from Telegram news comments, Kaspi and 2GIS reviews, each labeled by hand. I fine-tuned xlm-roberta-base as a three-way filter and put it up against FastText and Lingua on the same held-out gold test (n = 461).

LID model (gold test, n=461)	Macro-F1	Recall _{mixed}	Prec _{mixed}
FastText v2 (real + synthetic)	70.9%	49.1%	65.8%
Lingua v2 (token voting)	88.6%	98.8%	76.8%
XLM-R v2 (fine-tuned filter)	96.6%	95.0%	95.0%

Synthetic data doesn't transfer: a FastText model trained on 480,000 generated "mixed" lines scores **85% F1 on its own synthetic test** but drops to **63% macro-F1 on the real gold set**. Lingua reaches high recall by over-tagging mixed (precision 76.8%), which is the same failure mode at a smaller scale.

AT CORPUS SCALE

Run over a **331,468-message** corpus (Telegram, Kaspi, 2GIS across 10+ Kazakhstani cities), the filter keeps **16,364 (4.9%)** as real code-switching. That's roughly an order of magnitude below what the keyword heuristic claimed, which is the point: honest counts change what you can say about the corpus.

DOWNSTREAM TONE

On the filtered code-switched review slice, a binary tone head (XLM-R) hits **97.3% macro-F1** on a 525-example test. The set is small and the labels were LLM-drafted then audited, so I read it as evidence the filtered data is usable rather than as the headline result.

WHAT'S HONEST ABOUT IT

- The 16,364 "mixed" corpus labels are model-predicted; only the 3,076 gold examples are fully hand-checked.
- Tone labels started as LLM drafts and were then audited, so 97% partly measures agreement with that process.
- Solo capstone. The notebook grew across many sessions, so a clean single-pass rerun is still on the to-do list.

WHERE I'D LIKE INPUT

Two things I'm unsure about: whether the loanword-vs-switch distinction is framed the way the VarDial / CALCS communities would want, and which venue fits a single-author resource contribution (gold LID set, weights, and the labeling tool) best. Short reactions are welcome; a longer read of the repo is a bonus, not an ask.

Everything is open: [code](#) + [notebook](#), [weights](#) and [the gold dataset](#). Reach me at bogsav999@gmail.com.